

TCS NQT Paper 07

Numerical Ability

Question 1

– Determine the smallest number 300 and 400 which when divided by 2,

3, 4, 5, 6 leaves the remainder 1, 2, 3, 4, 5 respectively?

- 1. 349
- 2. 329
- 3. 339
- 4. 359

****Answer:** 359**

Question 2

– What is the average (nearest to integer) of all semi-primes between 20

And 40?

- 1. 30
- 2. 32
- 3. 31
- 4. 28

****Answer:** 30**

Question 3

– In a class, the ratio of the number of boys to girls is 8:7 and the average

weight of the boys is 20% less than the average weight of the girls. If the total weight of the class is 3417kg, then find the difference between the weight of all the girls and All the boys.

- 1. 143
- 2. 173
- 3. 163
- 4. 153

****Answer:** Option D – 153**

Question 4

– The sum of money is loaned at compound interest for 2 years at 20%

per annum. It would fetch Rs. 482 more, if the interest was payable half-yearly than if it were payable yearly, then the sum is:

- 1. 30000
- 2. 20000
- 3. 40000
- 4. 15000

****Answer:** 20000**

Question 5

– Find the compound interest on Rs. 3200 in 1.5 years at 5% per annum if

interest is calculated half-yearly.

- 1. Rs. 344.05
- 2. Rs. 3446.05
- 3. Rs. 346.05
- 4. Rs. 246.05

****Answer:** 246.05**

Question 6

– On an article $x\%$ commission is given on the retail price, cost price is

Rs. 60. After selling, the shopkeeper gets 12% profit. If commission is reduced by $y\%$ then profit percentage is 68%. What is the value of x and y ?

- 1. 40%, 30%
- 2. 30%, 40%
- 3. 40%, 20%
- 4. 20%, 40%

****Answer:**** 30%, 40%

Question 7

– A man has Rs. 13, 600. He lends some parts of his money at the rate of

10% simple interest and some part at 7.5% yearly. After 3.5 years he earned Rs. 3808. Find the money given at 10% rate.

- 1. Rs. 2,420
- 2. Rs. 2,780
- 3. Rs. 2,720
- 4. Rs. 5,600

****Answer:**** Rs. 2,720

Question 8

– Nitya divided an amount of Rs. 15750 among her four daughters Priya,

Pooja, Ritu and Rina in ratio 9 : 3 : 4 : 5 respectively for investing. At the end of the year the percentage profit of Priya was equal to the percentage loss of Pooja and the percentage loss of Ritu was equal to the percentage loss of Rina. Calculate the overall profit or loss for Nitya at the end of the year if the percentage loss suffered By Pooja was 12% and percentage loss suffered by Ritu was 6%.

- 1. Profit of Rs. 165
- 2. Loss of Rs. 165

- 3. Loss of Rs. 135
- 4. Profit of Rs. 135

****Answer:**** Profit of Rs. 135

Question 9

– What is the simplified value

of $(0.111 \times 1.3 - 1.5)0.571428 - (11 \times 1.3 - 1.50.1)0.571428$

- 1. 0.03125
- 2. 0.015625
- 3. 0.08
- 4. 0.0625

****Answer:**** 0.0625

Question 10

– The salary of Ram and Sujatha are in the ratio of 4 : 3. The salary of

Ram and Sujatha is increased by Rs. 3000 and the ratio changes to 9 : 8. Ram's salary after increase will be:

- 1. Rs. 4500
- 2. Rs. 8300
- 3. Rs. 5400

- 4. Rs. 3800

****Answer:**** Rs. 5400

Question 11

– If a is the rate per cent corresponding to the fraction $\frac{24}{25}$, and if 30

km is $b\%$ of 40 km, then what percentage (correct upto two decimal places) is b of A^2 ?

- 1. 0.81

- 2. 78.13

- 3. 7.81

- 4. 8.13

****Answer:**** 0.81

Question 12

– A car covers its half journey at a speed of 40 km per hour. Remaining

half at a speed of 60 km per hour. Total journey of 12 hrs Find the distance traveled By the car in Km.

- 1. 432

- 2. 144

- 3. 576

- 4. 288

****Answer:**** 576

Question 13

– A person travels the first part of his journey at 120 kmph and the second part at 180 kmph and covers the total distance of 2160 km to his destination in 15 hours. How long did the first part of his journey last?

- 1. 8 hours
- 2. 9 hours
- 3. 7 hours
- 4. 10 hours

****Answer:**** 9 hours

Question 14

– A car has tyre has two punctures. The first puncture alone would have made the tyre flat in 7 min and the second alone would have done it in 4 min. If air leaks out at a constant rate , how long does it take both the punctures together to make it flat.

- 1. 41114111
- 2. 26112116

- 3. 31113111

- 4. 16111116

****Answer:**** 26112116

Question 15

– If the cost of an article is 71% less than its selling price which is 12.5% less than its marked price, the discount offered is what percent of the profit gained.

****Answer:**** _____

Question 16

– The mean of 11 observations is 40. A new observation is added then finds the mean of 12 observations.

- 1. 38.4

- 2. 39.4

- 3. 37.4

- 4. 36.4

****Answer:**** Wrong question

Question 17

– A man deposit Rs. 600 for 3 years, 750 for 4 years at $r\%$ p.a. rate of

simple interest (S.I.) and Rs. 1200 for 6 years at $s\%$ p.a. He got S.S. Rs. 1200 total from all deposits. Find the $r + s$ if difference between r and s is equal to 5.

****Answer:**** $r+s=13$

Question 18

– The curved surface area of a right circular cylinder is 1320 cm^2 .

If the total surface area is 130% of the covered surface area, then find its Approximate radius (in cm) (use $\pi=22/7$).

- 1. 8.28

- 2. 9.04

- 3. 7.94

- 4. 7.62

****Answer:**** 7.94

Question 19

– Study the given table answer the question that follow

The table shows the number of people (in hundreds) participating in an annual fair from six different towns over the year.

What percentage of people are less than that of the total number of people participating in the fair from town D in the year 2016 and 2017 together with the number of people participating in the fair from town C in the same years?

****Answer:**** 20%

Question 20

– If $0 < a < 1$, $b > 2$ and $C < 0$, and $x = ab - cd \times c^2 a x = ba - dc \times a c^2$, then which of the following is correct?

- 1. $X < 1$ always
- 2. $X < 0$
- 3. $X > 1$ always
- 4. $X > 0$

****Answer:**** $x < 1$ always

Verbal Ability

Question 1

– Parts of the sentence have been given as options. One of them

Contains a grammatical error. Select the option that has the error.

Developing the ability to understand and use nonverbal communication can help you

Connect with others, express what you really mean, navigate challenging situations,

And build much more better relationships at home and work.

- 1. Developing the ability to understand and use non verbal.
- 2. Build much more better relationships at home and work
- 3. Really mean, navigate challenging situations and
- 4. Communication can help you connect with others, express what you.

****Answer:**** Build much more better relationships at home and work

Question 2

– Select the sentence in which the underlined idiom has been used

Appropriately.

1. The minister's popularity among elderly voters gives him an ace on his sleeve for the Coming election in his native city.

- 2. The Indian cricket had an ace upon their sleeve in the form of Keerti Shankar, the budding new all-rounder.

- 3. Sugandha kept quiet at the board meeting, who knew she had an ace hidden up her sleeve all this while.

- 4. The politician's rival had a sleeve up to his ace in the form of a scandalous secret that he revealed in a completely public gathering.

****Answer:**** Sugandha kept quiet at the board meeting, who knew she had an ace hidden up

her sleeve all this while.

Question 3

– Parts of the given sentence have been given as options. One of them

Contains a grammatical error. Select the option that has the error.

The UK economy had been one of the strongest economy in terms of inflation, interest rates and unemployment which remained low until the recession.

- in terms of inflation
- one of the strongest economy
- which remained low until the recession.
- The UK economy had been

****Answer:**** one of the strongest economy

Question 4

– The pandemic has given unprecedented rise to the 1.

for working from home. AND people are allured by the opportunity which 2.

Any chance of commuting in heavy traffic. However, this 3. is not devoid of challenges and difficulties. In a recently conducted survey, 41% of 4. working from home reported burned out.

Blank 1. A. requirement B. claim C. interest D. demand Blank 2. Prevents B. hinders C. stops D. presents Blank 3. A. schedule B. arrangement C. package D. deal Blank 4. A. employers B. women C. employees D. students

Question 5

– The sentence contains four underlined phrases. Three of these phrases

correctly used and one has been incorrectly used. Select the phrase which best replaces the incorrect phrase in the sentence.

Secularism is the need for the hour for uplifting the nation from the abyss of religious myths and beliefs, waging a united war against the poison of communalism And other social evils.

- 1. Need of the hour
- 2. An abyss in
- 3. Across the
- 4. Also to wage

****Answer:**** Need of the hour

Question 6

– In the following passage, some idioms have been deleted. Read the

passage carefully and select the most appropriate option to fill in each blank.

How to live a reasonably happy and comfortable life is 1. One can live a happy and satisfied life despite 2. at us at times. The first and foremost thing to do is to find people that you like and who won't make you feel 3. because that is a terrible feeling. Its also a good idea to find people who aren't going to 4. too much. Friends do kid around but good friends hit a happy medium between joking and respecting each other.

- 1. 1. no great rocket science 2. the ups and down of life 3. let down 4. push the buttons

- 2. 1. no great adventure 2. the lines that life throws 3. grapple with 4. put up with
- 3. 1. no great rocket science 2. the curves that life throws 3. put down 4. press your buttons
- 4. 1. is no uphill task 2. the twists and turns that life throws 3. put off 4. try your patience

****Answer:**** 1. no great rocket science 2. the curves that life throws 3. put down 4. press your buttons

Question 7

– In the following passage, some idioms have been deleted. Read the

passage carefully and select the most appropriate option to fill in each blank.

We all love buying the latest, and the most sophisticated products such as smart and stylish clothing. But, we must remember that 1. . It pays to think with a cool mind when shopping. Instead of 2. , of using your credit card on an impulse it would be better to postpone the shopping for a day or two. Try doing this the next time your 3. because some beautiful object is beckoning to you from the shop window. Once you've got this technique 4 , you'd be surprised how much you will save.

- 1. 1. all that glitters is not gold 2. falling into the trap 3. heart skips a beat 4. under your belt
- 2. 1. all that glitter and glitterati 2. falling into the quicksand 3. heart misses a beat 4. under your pants
- 3. 1. all that glitters is not gold 2. falling into the rat race 3. heartbeat goes up 4. under your nose
- 4. 1. all that glistens is not gold 2. falling the quagmire 3. heart misses beat 4. under your Charge

****Answer:**** 1. all that glitters is not gold 2. falling into the trap 3. heart skips a beat

4. under your belt

Question 8

– Sentences of a paragraph are given below in jumbled order. Arrange the sentences in correct order to form a coherent and meaningful passage.

1. Like elephants, a manatee's molars wear down and move forward and fall out, and are replaced by new teeth coming in at the back of the mouth.
2. While most mammals— from giraffes to cats— have seven neck vertebrae, only tree sloths and manatees defy that rule.
3. A manatee has only six vertebrae in its neck.
4. They diverged from common land mammal over 50 million years ago
5. The closest living relatives to the manatee are the elephant and the hyrax.

Options.

- 1. ADEBC
- 2. EADBC
- 3. ABCDE
- 4. AEBCD

****Answer:**** AEBCD

Question 9

– In the following passage, some words have been deleted. Read the

passage carefully and select the most appropriate option to fill in each blank.

While there is no denying that the world loves a winner, it is important that you

A. the signs of stress in your behavior and be healthy enough to enjoy your B.

. Stress can strike anytime, in a fashion that may leave you unaware of its

C. in your life. While a certain amount of pressure is necessary for D. , it

is important to be able to recognize your individual limit. There are some individuals who accept competition in a healthy fashion. There are others who collapse under pressure.

- 1. (A) notice (B) knowledge (C) company (D) conduct

- 2. (A) verify (B) progress (C) absence (D) appearance

- 3. (A) remember (B) benefit (C) attendance (D) completion

- 4. (A) recognize (B) success (C) presence (D) performance

****Answer:**** (A) recognize (B) success (C) presence (D) performance

Question 10

– Four idioms are given below. Choose their sequence that would fill in

the blanks and complete the text given here.

Along with the traditional golden watch and the retirement dinner speeches

were the highlights of Mr. Rath's party. The company had been seeing a lot of

employees either being fired or quitting, so it was a nice change to see everyone in a

good mood. In the previous week, his younger brother was ejected from the same Company but not without . Another employee was retained from offering him , which he took as .

1. A golden opportunity
2. A golden handshake
3. A golden parachute
4. Golden handcuffs

- 1. 3124
- 2. 3421
- 3. 2143
- 4. 2341

****Answer:** 2341**

Question 11

– Many mental health conditions can be effectively treated , yet

the gap between people needing care and those with access to care remains substantial. Effective treatment low.

- 1. at a relative low cost; coverage remains extreme
- 2. at a relatively a low cost, coverage remains extremely
- 3. At a relatively low cost; coverage remains extremely
- 4. at relatively a low cost; coverage extremely remains

****Answer:**** At a relatively low cost; coverage remains extremely

Question 12

– Select the most appropriate option that can substitute the underlined

words in the given sentence.

A large group of college students were gathered around the voting booth.

- 1. Congregation
- 2. Herd
- 3. Flock
- 4. Horde

****Answer:**** horde

Question 13

– Sentences of a paragraph are given below in jumbled order. Arrange

the order in correct order to form a coherent and meaningful passage.

1. Women tend to have less severe OSA than males, with a lower apnea-hypopnea index (HI) and shorter apneas and hypopneas.
2. However, the importance of ISA in women is increasingly being recognized, along with A number of significant gender-related differences in the symptoms, diagnosis, Consequences, and treatment of OSA.
3. Obstructive sleep apnea (OSA) has traditionally been seen as a male disease.
4. Episodes of upper airway resistance that do not meet the criteria for apneas are more

Common in women. Prevalence rates are lower in women, and proportionally fewer women receive a correct diagnosis.

5. Research has also documented sex differences in the upper airway, fat distribution, and respiratory stability in OSA.

- 1. CBADE
- 2. ADCEB
- 3. BCADE
- 4. ABCDE

****Answer:**** CBADE

Question 14

– Select the most appropriate option to fill in the blanks

We all want to be successful, but we one another about what this means or how we should achieve it. Office politics personality and opinion become Difficult to manage.

- 1. Don't always agrees with; arises when the differences of
- 2. Don't always agree to; arise when those differences in
- 3. Don't always agree with; arise when these differences of
- 4. Doesn't always agree with; arises when these differences of

****Answer:**** Don't always agree with; arise when these differences of

Question 15

– For the four-sentence (S1 to S4) in the paragraph below, sentences S1

and S4 are given. From the options P, Q, R, and S choose the appropriate sentences for S2 and S3, respectively.

S1. The Constitution of India guarantees the protection of life and personal liberty to one and all.

S2.

S3.

S4. One can also get information about the Supreme Court, High Courts, Subordinate Courts, Legal Aid, Profession, Alternative Dispute Resolution, etc.

1. There was a great opportunity to gain inside access to the grand interiors of the Parliament House building.
2. It provides adequate safeguards to fundamental rights against arbitrary decisions.
3. Citizens also have the right to detailed information about various legislations, rules, regulations, etc.
4. The District Courts of India are the courts established by the state governments in India for every district.

- 1. PQ

- 2. SQ

- 3. RQ

- 4. RP

****Answer:** RQ**

Question 16

– For the four- sentence paragraph below, sentences S1 and S4 are

given. From P, Q, R and S, select the appropriate sentences that can fit S2 and S3, respectively.

S1: Charlie Chaplin was one of the greatest and widely loved silent movie stars.

S2:

S3:

S4: The little tramp, a well-meaning man in a raggedy suit with cane, always found himself wobbling into awkward situations and miraculously wobbling away.

P: From “Easy street” (1971) to “Modern Times” (1936), he made many of the funniest and most popular films of his time.

Q: He was best known for his character, the naive and lovable Little Tramp.

R: More than any other figure, it is this character played by Charlie Chaplin that we Associate with the time before the talkies.

S: The same year he made thirty-four short films.

****Answer:**** QR

Question 17

– Given below is a paragraph with five sentences, which are presented

in a jumbled order. Write down the best order (e.g. ABCDE) which would form a meaningful, unified and coherent paragraph

- 1. Connected citizens will find themselves empowered to some extent in novel ways, but this empowerment will come with serious downsides – for example in the erosion – perhaps elimination – of privacy.
- 2. A digital caste system will endure well into the future, and the extent to which people will benefit from the technology will be critically dependent on their positions in that system
- 3. Cyberspace will not “overtake or overhaul” the existing world order, but it will make everything much more complicated.
- 4. States will find that they will need domestic policies to deal with both the physical and virtual worlds, and also two foreign policies – one for “normal” international relations, And one for cyberspace– the policies will be different in each case. 5. Poor people will be the biggest beneficiaries simply because of where they live, but they will also face the worst drawbacks of the digital age.

****Answer:**** BECDA

Question 18

– Read the given passage and answer the questions that follow.

In contrast to traditional analyses of minority business, the sociology analysis contends that minority business ownership is a group-level phenomenon, in that it is largely dependent upon social-group resources for its development. Specifically, the Analysis indicates that support networks play a critical role in starting and maintaining minority business enterprises by providing owners with a range of Assistance, from the informal encouragement of family members and friends to dependable sources of labor and clientele from the owner’s ethnic group. Such self-

help networks, which encourage and support ethnic minority entrepreneurs, consist of 'primary' institutions, those closest to the individual in shaping his or her behavior and beliefs. They are characterized by the face-to-face association and cooperation of persons united by ties of mutual concern. They form an intermediate social level between the individual and larger 'secondary' institutions based on impersonal relationships. Primary institutions comprising the support network include kinship, peer and neighborhood or community subgroups.

A major function of self-help networks is financial support. Most scholars agree that minority business owners have depended primarily on family funds and ethnic community resources for investment capital. Personal savings have been accumulated, often through frugal living habits that require sacrifices from the entire family and thus as a product of long term family financial behavior. Additional loans and gifts from relatives, forthcoming because of group obligation rather than narrow investment calculation, have supplemented personal savings. Individual entrepreneurs do not necessarily rely on their kin because they cannot obtain

financial backing from commercial resources. They may actually avoid banks because they assume commercial institutions either cannot comprehend the special needs of minority enterprises or charge unreasonably high interest rates.

Within the larger ethnic community, rotating credit associations have been used to raise capital. These associations are informal clubs of friends and other trusted members of the ethnic group who make regular contributions to a fund that is given to each contributor in rotation. One author estimates that 40 percent of New York Chinatown firms established during 1900-1950 utilized such associations as their initial source of capital. However, recent immigrants and third or fourth generations of older groups now employ rotating credit associations only occasionally to raise investment funds. Some groups, like Black Americans, found other means of

financial support for their entrepreneurial efforts. The first Black-operated banks were created in the late nineteenth century as depositories for dues collected from fraternal or lodge groups, which themselves had sprung from Black churches. Black Banks made limited investments in other Black enterprises. Irish immigrants in American cities organized many building and loan associations to provide capital for home construction and purchase. They, in turn, provided work for many Irish home-Building contractor firms. Other ethnic and minority groups followed similar practices in founding ethnic-directed financial institutions.

Why do the traditional analysis not contend as opposed to the sociological analysis?

- 1. Minority run businesses have access to banking finance.
- 2. Special government funding is also available to minority run businesses, as per the traditional school
- 3. Kinship communities lend to minorities' businesses only
- 4. Institutional finance is keen on lending to minority run businesses

****Answer:**** Minority run businesses have access to banking finance.

Question 19

– Read the given passage and answer the questions that follow.

In contrast to traditional analyses of minority business, the sociology analysis contends that minority business ownership is a group-level phenomenon, in that it is largely dependent upon social-group resources for its development. Specifically, the Analysis indicates that support networks play a critical role in starting and maintaining minority business enterprises by providing owners with a range of Assistance, from the informal encouragement of family members and friends to

dependable sources of labor and clientele from the owner's ethnic group. Such self-help networks, which encourage and support ethnic minority entrepreneurs, consist of 'primary' institutions, those closest to the individual in shaping his or her behavior and beliefs. They are characterized by the face-to-face association and cooperation of persons united by ties of mutual concern. They form an intermediate social level between the individual and larger 'secondary' institutions based on impersonal relationships. Primary institutions comprising the support network include kinship, peer and neighborhood or community subgroups.

A major function of self-help networks is financial support. Most scholars agree that minority business owners have depended primarily on family funds and ethnic community resources for investment capital. Personal savings have been accumulated, often through frugal living habits that require sacrifices from the entire family and thus as a product of long term family financial behavior. Additional loans and gifts from relatives, forthcoming because of group obligation rather than narrow investment calculation, have supplemented personal savings. Individual entrepreneurs do not necessarily rely on their kin because they cannot obtain financial backing from commercial resources. They may actually avoid banks because they assume commercial institutions either cannot comprehend the special needs of minority enterprise or charge unreasonably high interest rates.

Within the larger ethnic community, rotating credit associations have been used to raise capital. These associations are informal clubs of friends and other trusted members of the ethnic group who make regular contributions to a fund that is given to each contributor in rotation. One author estimates that 40 percent of New York Chinatown firms established during 1900-1950 utilized such associations as their initial source of capital. However, recent immigrants and third or fourth generations

of older groups now employ rotating credit associations only occasionally to raise investment funds. Some groups, like Black Americans, found other means of

financial support for their entrepreneurial efforts. The first Black-operated banks were created in the late nineteenth century as depositories for dues collected from fraternal or lodge groups, which themselves had sprung from Black churches. Black Banks made limited investments in other Black enterprises. Irish immigrants in American cities organized many building and loan associations to provide capital for home construction and purchase. They, in turn, provided work for many Irish home-Building contractor firms. Other ethnic and minority groups followed similar practices in founding ethnic-directed financial institutions.

According to sociological analysis, the following is not true

- 1. Secondary institutions fund ethnic groups regularly
- 2. Black banks are offshoot of the same community lending ethos
- 3. It has its roots in ethnic bias
- 4. It refers to a class action movement to fund one's own.

****Answer:**** It refers to a class action movement to fund one's own.

Question 20

– Read the given passage and answer the questions that follow.

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The sociological findings do not prove following

- 1. Communities are more assured when they deal with people from their own race or emigrant nations
- 2. There is rampant ethnic and racial bias in the investment sector
- 3. Social hyper cross pollination had not begun at the time of these studies
- 4. Ways and means of approaching resources are limited for marginal groups.

****Answer:** C**

Question 21

– Read the given passage and answer the questions that follow

A small pole about thirty centimeters long stuck up from a square box on the lab table. There was a ring, only a few centimeters across, encircling the base of the pole. The strange item would have looked a bit like a carnival ring-toss game were it not for the cord snaking out of the side of the box and the single toggle switch on top.

“What the heck’s this?” Kusanagi muttered, bending down to stare at the device.

“I wouldn’t touch that if I were you,” warned Kishitani beside him.

“Nah, I’m sure it’s fine. If this thing were dangerous, he never would’ve left it lying out here like

this.” Kusanagi flicked the switch. The moment he did so, the ring around the pole began to float upward.

Kusanagi gaped. The ring was hovering in mid-air, wobbling slightly.

“Try pushing the ring down,” came a voice from behind him.

Kusanagi looked around as Yukawa walked into the room, a file folder cradled in his arms

“Welcome back. Class?” Kusanagi asked as he turned back to the floating ring. Reaching out

gingerly, he tried to push it down with his fingertips. A second later he yanked back his hand.

“Yowch! That thing’s hot!”

“It’s true, I’m not in the habit of leaving dangerous objects lying around. I do, however, assume a

basic knowledge of physics.” Yukawa strolled over to the table and flicked off the switch. “This

Device is only high school-level physics, as a matter of fact.”

“Well, I didn’t take physics in high school,” Kusanagi said, blowing on his fingertips.

Kishitani

laughed immoderately.

“Who’s your friend?” Yukawa asked with a raised eyebrow.

The smile vanished from Kishitani’s face and he bowed curtly. “Kishitani. I work with Detective

Kusanagi. I’ve heard a lot about you, Professor. You’ve helped us out with a lot of Cases, haven’t you? They call you Detective Galileo down at the station.”

Yukawa frowned and waved his hand. “Don’t call me that. I don’t help out because I enjoy it, you

know. I just couldn’t bear listening to Detective Kusanagi’s faulty attempts at reasoning on a certain occasion, and I made the mistake of correcting him. I’d be wary of spending too much time with him, if I were you. You might catch whatever he has that’s hardened his brain into a rock.”

Which option gives an erroneous description of the object that Kusanagi and Kishitani found in Yukawa’s lab?

1. The device or the experiment worked only when the single toggle switch on top would be turned on.
2. The device, based on experimentation related to concepts of Physics, had a pole and a Box attached to it.

- 3. The square box on top of the thirty centimeters long pole had a cord attached to one side of it.
- 4. On the base of the pole, a ring encircled the base, which would float upwards if the switch was turned on.

****Answer:**** The square box on top of the thirty centimeters long pole had a cord attached to one side

of it.

Question 22

– On the basis of the narrative style and tone of the passage, what may

Be deduced as the most correct option?

1. The detectives had high regard and admiration for Yukawa's knowledge and even though he was not a part of the official police force, they did not hesitate to take his advice on Cases from time to time.

- 2. Other than Kusanagi and Kishitani, no one else in the police force knew about Yukawa's involvement in official cases and so, the two wanted to keep it a secret from others.

- 3. Professor Yukawa was a scientist who had high regard for detective work and therefore, whenever he was approached by the official police force to help them in any case, he obliged

- 4. Professor Yukawa encourages Kishitani to follow the principles of detective work of Kusanagi who had become a hardened detective as a result of his experience of several years of work

****Answer:**** Professor Yukawa was a scientist who had high regard for detective work

and therefore, whenever he was approached by the official police force to help them in any case, he obliged

Verbal Ability

Question 1

– Parts of the sentence have been given as options. One of them

Contains a grammatical error. Select the option that has the error.

Developing the ability to understand and use nonverbal communication can help you

Connect with others, express what you really mean, navigate challenging situations,

And build much more better relationships at home and work.

- 1. Developing the ability to understand and use non verbal.
- 2. Build much more better relationships at home and work
- 3. Really mean, navigate challenging situations and
- 4. Communication can help you connect with others, express what you.

****Answer:**** Build much more better relationships at home and work

Question 2

– Select the sentence in which the underlined idiom has been used

Appropriately.

1. The minister's popularity among elderly voters gives him an ace on his sleeve for the

Coming election in his native city.

- 2. The Indian cricket had an ace upon their sleeve in the form of Keerti Shankar, the budding new all-rounder.
- 3. Sugandha kept quiet at the board meeting, who knew she had an ace hidden up her sleeve all this while.
- 4. The politician's rival had a sleeve up to his ace in the form of a scandalous secret that he revealed in a completely public gathering.

****Answer:**** Sugandha kept quiet at the board meeting, who knew she had an ace hidden up

her sleeve all this while.

Question 3

– Parts of the given sentence have been given as options. One of them

Contains a grammatical error. Select the option that has the error.

The UK economy had been one of the strongest economy in terms of inflation, interest rates and unemployment which remained low until the recession.

- in terms of inflation
- one of the strongest economy
- which remained low until the recession.
- The UK economy had been

****Answer:**** one of the strongest economy

Question 4

– The pandemic has given unprecedented rise to the 1.

for working from home. AND people are allured by the opportunity which 2.

Any chance of commuting in heavy traffic. However, this 3. is not devoid of challenges and difficulties. In a recently conducted survey, 41% of 4. working from home reported burned out.

Blank 1. A. requirement B. claim C. interest D. demand Blank 2. Prevents B. hinders C. stops D. presents Blank 3. A. schedule B. arrangement C. package D. deal Blank 4. A. employers B. women C. employees D. students

Question 5

– The sentence contains four underlined phrases. Three of these phrases

correctly used and one has been incorrectly used. Select the phrase which best replaces the incorrect phrase in the sentence.

Secularism is the need for the hour for uplifting the nation from the abyss of religious myths and beliefs, waging a united war against the poison of communalism And other social evils.

- 1. Need of the hour
- 2. An abyss in
- 3. Across the
- 4. Also to wage

****Answer:**** Need of the hour

Question 6

– In the following passage, some idioms have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank.

How to live a reasonably happy and comfortable life is 1. One can live a happy and satisfied life despite 2. at us at times. The first and foremost thing to do is to find people that you like and who won't make you feel 3. because that is a terrible feeling. Its also a good idea to find people who aren't going to 4. too much. Friends do kid around but good friends hit a happy medium between joking and respecting each other.

- 1. 1. no great rocket science 2. the ups and down of life 3. let down 4. push the buttons
- 2. 1. no great adventure 2. the lines that life throws 3. grapple with 4. put up with
- 3. 1. no great rocket science 2. the curves that life throws 3. put down 4. press your buttons
- 4. 1. is no uphill task 2. the twists and turns that life throws 3. put off 4. try your patience

****Answer:**** 1. no great rocket science 2. the curves that life throws 3. put down 4. press your buttons

Question 7

– In the following passage, some idioms have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank.

We all love buying the latest, and the most sophisticated products such as smart and

stylish clothing. But, we must remember that 1. . It pays to think with a cool mind when shopping. Instead of 2. , of using your credit card on an impulse it would be better to postpone the shopping for a day or two. Try doing this the next time your 3. because some beautiful object is beckoning to you from the shop window. Once you've got this technique 4 , you'd be surprised how much you will save.

- 1. 1. all that glitters is not gold 2. falling into the trap 3. heart skips a beat 4. under your belt
- 2. 1. all that glitter and glitterati 2. falling into the quicksand 3. heart misses a beat 4. under your pants
- 3. 1. all that glitters is not gold 2. falling into the rat race 3. heartbeat goes up 4. under your nose
- 4. 1. all that glistens is not gold 2. falling the quagmire 3. heart misses beat 4. under your Charge

****Answer:**** 1. all that glitters is not gold 2. falling into the trap 3. heart skips a beat

4. under your belt

Question 8

– Sentences of a paragraph are given below in jumbled order. Arrange the

sentences in correct order to form a coherent and meaningful passage.

1. Like elephants, a manatee's molars wear down and move forward and fall out, and are replaced by new teeth coming in at the back of the mouth.
2. While most mammals— from giraffes to cats— have seven neck vertebrae, only tree sloths and manatees defy that rule.

3. A manatee has only six vertebrae in its neck.
4. They diverged from common land mammal over 50 million years ago
5. The closest living relatives to the manatee are the elephant and the hyrax.

Options.

- 1. ADEBC
- 2. EADBC
- 3. ABCDE
- 4. AEBCD

****Answer:**** AEBCD

Question 9

– In the following passage, some words have been deleted. Read the

passage carefully and select the most appropriate option to fill in each blank.

While there is no denying that the world loves a winner, it is important that you

A. the signs of stress in your behavior and be healthy enough to enjoy your B.

. Stress can strike anytime, in a fashion that may leave you unaware of its

C. in your life. While a certain amount of pressure is necessary for D. , it

is important to be able to recognize your individual limit. There are some individuals who accept competition in a healthy fashion. There are others who collapse under pressure.

- 1. (A) notice (B) knowledge (C) company (D) conduct
- 2. (A) verify (B) progress (C) absence (D) appearance

- 3. (A) remember (B) benefit (C) attendance (D) completion
- 4. (A) recognize (B) success (C) presence (D) performance

****Answer:**** (A) recognize (B) success (C) presence (D) performance

Question 10

– Four idioms are given below. Choose their sequence that would fill in

the blanks and complete the text given here.

Along with the traditional golden watch and the retirement dinner speeches were the highlights of Mr. Rathi's party. The company had been seeing a lot of employees either being fired or quitting, so it was a nice change to see everyone in a good mood. In the previous week, his younger brother was ejected from the same Company but not without . Another employee was retained from offering him , which he took as .

1. A golden opportunity
2. A golden handshake
3. A golden parachute
4. Golden handcuffs

- 1. 3124
- 2. 3421
- 3. 2143
- 4. 2341

****Answer:**** 2341

Question 11

– Many mental health conditions can be effectively treated , yet

the gap between people needing care and those with access to care remains substantial. Effective treatment low.

- 1. at a relative low cost; coverage remains extreme
- 2. at a relatively a low cost, coverage remains extremely
- 3. At a relatively low cost; coverage remains extremely
- 4. at relatively a low cost; coverage extremely remains

****Answer:**** At a relatively low cost; coverage remains extremely

Question 12

– Select the most appropriate option that can substitute the underlined

words in the given sentence.

A large group of college students were gathered around the voting booth.

- 1. Congregation
- 2. Herd
- 3. Flock
- 4. Horde

****Answer:**** horde

Question 13

– Sentences of a paragraph are given below in jumbled order. Arrange

the order in correct order to form a coherent and meaningful passage.

1. Women tend to have less severe OSA than males, with a lower apnea-hypopnea index (HI) and shorter apneas and hypopneas.
2. However, the importance of ISA in women is increasingly being recognized, along with A number of significant gender-related differences in the symptoms, diagnosis, Consequences, and treatment of OSA.
3. Obstructive sleep apnea (OSA) has traditionally been seen as a male disease.
4. Episodes of upper airway resistance that do not meet the criteria for apneas are more Common in women. Prevalence rates are lower in women, and proportionally fewer women receive a correct diagnosis.
5. Research has also documented sex differences in the upper airway, fat distribution, and respiratory stability in OSA.

- 1. CBADE

- 2. ADCEB

- 3. BCADE

- 4. ABCDE

****Answer:**** CBADE

Question 14

– Select the most appropriate option to fill in the blanks

We all want to be successful, but we one another about what this means or how we should achieve it. Office politics personality and opinion become Difficult to manage.

- 1. Don't always agrees with; arises when the differences of
- 2. Don't always agree to; arise when those differences in
- 3. Don't always agree with; arise when these differences of
- 4. Doesn't always agree with; arises when these differences of

****Answer:**** Don't always agree with; arise when these differences of

Question 15

– For the four-sentence (S1 to S4) in the paragraph below, sentences S1

and S4 are given. From the options P, Q, R, and S choose the appropriate sentences for S2 and S3, respectively.

S1. The Constitution of India guarantees the protection of life and personal liberty to one and all.

S2.

S3.

S4. One can also get information about the Supreme Court, High Courts, Subordinate Courts, Legal Aid, Profession, Alternative Dispute Resolution, etc.

1. There was a great opportunity to gain inside access to the grand interiors of the

Parliament House building.

2. It provides adequate safeguards to fundamental rights against arbitrary decisions.
3. Citizens also have the right to detailed information about various legislations, rules, regulations, etc.
4. The District Courts of India are the courts established by the state governments in India for every district.

- 1. PQ
- 2. SQ
- 3. RQ
- 4. RP

****Answer:**** RQ

Question 16

– For the four- sentence paragraph below, sentences S1 and S4 are

given. From P, Q, R and S, select the appropriate sentences that can fit S2 and S3, respectively.

S1: Charlie Chaplin was one of the greatest and widely loved silent movie stars.

S2:

S3:

S4: The little tramp, a well-meaning man in a raggedy suit with cane, always found himself wobbling into awkward situations and miraculously wobbling away.

P: From “Easy street” (1971) to “Modern Times” (1936), he made many of the funniest and most popular films of his time.

Q: He was best known for his character, the naive and lovable Little Tramp.

R: More than any other figure, it is this character played by Charlie Chaplin that we Associate with the time before the talkies.

S: The same year he made thirty-four short films.

****Answer:**** QR

Question 17

– Given below is a paragraph with five sentences, which are presented

in a jumbled order. Write down the best order (e.g. ABCDE) which would form a meaningful, unified and coherent paragraph

- 1. Connected citizens will find themselves empowered to some extent in novel ways, but this empowerment will come with serious downsides – for example in the erosion – perhaps elimination – of privacy.
- 2. A digital caste system will endure well into the future, and the extent to which people will benefit from the technology will be critically dependent on their positions in that system
- 3. Cyberspace will not “overtake or overhaul” the existing world order, but it will make everything much more complicated.
- 4. States will find that they will need domestic policies to deal with both the physical and virtual worlds, and also two foreign policies – one for “normal” international relations, And one for cyberspace– the policies will be different in each case.
- 5. Poor people will be the biggest beneficiaries simply because of where they live, but they will also face the worst drawbacks of the digital age.

****Answer:**** BECDA

Question 18

– Read the given passage and answer the questions that follow.

In contrast to traditional analyses of minority business, the sociology analysis contends that minority business ownership is a group-level phenomenon, in that it is largely dependent upon social-group resources for its development. Specifically, the Analysis indicates that support networks play a critical role in starting and maintaining minority business enterprises by providing owners with a range of Assistance, from the informal encouragement of family members and friends to dependable sources of labor and clientele from the owner's ethnic group. Such self-help networks, which encourage and support ethnic minority entrepreneurs, consist

of 'primary' institutions, those closest to the individual in shaping his or her behavior And beliefs. They are characterized by the face-to-face association and cooperation of persons united by ties of mutual concern. They form an intermediate social level Between the individual and larger 'secondary' institutions based on impersonal relationships. Primary institutions comprising the support network include kinship, peer and neighborhood or community subgroups.

A major function of self-help networks is financial support. Most scholars agree that minority business owners have depended primarily on family funds and ethnic Community resources for investment capital. Personal savings have been Accumulated, often through frugal living habits that require sacrifices from the entire family and thus as a product of long term family financial behavior. Additional loans And gifts from relatives, forthcoming because of group obligation rather than narrow investment calculation, have supplemented personal savings. Individual entrepreneurs do not necessarily rely on their kin because they cannot obtain

financial backing from commercial resources. They may actually avoid banks because they assume commercial institutions either cannot comprehend the special needs of minority enterprises or charge unreasonably high interest rates.

Within the larger ethnic community, rotating credit associations have been used to raise capital. These associations are informal clubs of friends and other trusted members of the ethnic group who make regular contributions to a fund that is given to each contributor in rotation. One author estimates that 40 percent of New York Chinatown firms established during 1900-1950 utilized such associations as their initial source of capital. However, recent immigrants and third or fourth generations of older groups now employ rotating credit associations only occasionally to raise investment funds. Some groups, like Black Americans, found other means of financial support for their entrepreneurial efforts. The first Black-operated banks were created in the late nineteenth century as depositories for dues collected from fraternal or lodge groups, which themselves had sprung from Black churches. Black Banks made limited investments in other Black enterprises. Irish immigrants in American cities organized many building and loan associations to provide capital for home construction and purchase. They, in turn, provided work for many Irish home-building contractor firms. Other ethnic and minority groups followed similar practices in founding ethnic-directed financial institutions.

Why do the traditional analysis not contend as opposed to the sociological analysis?

- 1. Minority run businesses have access to banking finance.
- 2. Special government funding is also available to minority run businesses, as per the traditional school
- 3. Kinship communities lend to minorities' businesses only
- 4. Institutional finance is keen on lending to minority run businesses

****Answer:**** Minority run businesses have access to banking finance.

Question 19

– Read the given passage and answer the questions that follow.

In contrast to traditional analyses of minority business, the sociology analysis contends that minority business ownership is a group-level phenomenon, in that it is largely dependent upon social-group resources for its development. Specifically, the Analysis indicates that support networks play a critical role in starting and maintaining minority business enterprises by providing owners with a range of Assistance, from the informal encouragement of family members and friends to dependable sources of labor and clientele from the owner's ethnic group. Such self-help networks, which encourage and support ethnic minority entrepreneurs, consist of 'primary' institutions, those closest to the individual in shaping his or her behavior And beliefs. They are characterized by the face-to-face association and cooperation of persons united by ties of mutual concern. They form an intermediate social level Between the individual and larger 'secondary' institutions based on impersonal relationships. Primary institutions comprising the support network include kinship, peer and neighborhood or community subgroups.

A major function of self-help networks is financial support. Most scholars agree that minority business owners have depended primarily on family funds and ethnic Community resources for investment capital. Personal savings have been

Accumulated, often through frugal living habits that require sacrifices from the entire family and thus as a product of long term family financial behavior. Additional loans And gifts from relatives, forthcoming because of group obligation rather than narrow investment calculation, have supplemented personal savings. Individual

entrepreneurs do not necessarily rely on their kin because they cannot obtain financial backing from commercial resources. They may actually avoid banks because they assume commercial institutions either cannot comprehend the special needs of minority enterprise or charge unreasonably high interest rates.

Within the larger ethnic community, rotating credit associations have been used to raise capital. These associations are informal clubs of friends and other trusted members of the ethnic group who make regular contributions to a fund that is given to each contributor in rotation. One author estimates that 40 percent of New York Chinatown firms established during 1900-1950 utilized such associations as their initial source of capital. However, recent immigrants and third or fourth generations of older groups now employ rotating credit associations only occasionally to raise investment funds. Some groups, like Black Americans, found other means of financial support for their entrepreneurial efforts. The first Black-operated banks were created in the late nineteenth century as depositories for dues collected from fraternal or lodge groups, which themselves had sprung from Black churches. Black Banks made limited investments in other Black enterprises. Irish immigrants in American cities organized many building and loan associations to provide capital for home construction and purchase. They, in turn, provided work for many Irish home-building contractor firms. Other ethnic and minority groups followed similar practices in founding ethnic-directed financial institutions.

According to sociological analysis, the following is not true

- 1. Secondary institutions fund ethnic groups regularly
- 2. Black banks are offshoot of the same community lending ethos
- 3. It has its roots in ethnic bias
- 4. It refers to a class action movement to fund one's own.

****Answer:**** It refers to a class action movement to fund one's own.

Question 20

– Read the given passage and answer the questions that follow.

In contrast to traditional analyses of minority business, the sociology analysis contends that minority business ownership is a group-level phenomenon, in that it is largely dependent upon social-group resources for its development. Specifically, the Analysis indicates that support networks play a critical role in starting and maintaining minority business enterprises by providing owners with a range of Assistance, from the informal encouragement of family members and friends to dependable sources of labor and clientele from the owner's ethnic group. Such self-help networks, which encourage and support ethnic minority entrepreneurs, consist of 'primary' institutions, those closest to the individual in shaping his or her behavior And beliefs. They are characterized by the face-to-face association and cooperation of persons united by ties of mutual concern. They form an intermediate social level Between the individual and larger 'secondary' institutions based on impersonal relationships. Primary institutions comprising the support network include kinship, peer and neighborhood or community subgroups.

A major function of self-help networks is financial support. Most scholars agree that minority business owners have depended primarily on family funds and ethnic Community resources for investment capital. Personal savings have been Accumulated, often through frugal living habits that require sacrifices from the entire family and thus as a product of long term family financial behavior. Additional loans And gifts from relatives, forthcoming because of group obligation rather than narrow investment calculation, have supplemented personal savings. Individual entrepreneurs do not necessarily rely on their kin because they cannot obtain financial backing from commercial resources. They may actually avoid banks

Because they assume commercial institutions either cannot comprehend the special needs of minority enterprise or charge unreasonably high interest rates.

Within the larger ethnic community, rotating credit associations have been used to raise capital. These associations are informal clubs of friends and other trusted members of the ethnic group who make regular contributions to a fund that is given to each contributor in rotation. One author estimates that 40 percent of New York

Chinatown firms established during 1900-1950 utilized such associations as their initial source of capital. However, recent immigrants and third or fourth generations of older groups now employ rotating credit associations only occasionally to raise investment funds. Some groups, like Black Americans, found other means of financial support for their entrepreneurial efforts. The first Black-operated banks were created in the late nineteenth century as depositories for dues collected from fraternal or lodge groups, which themselves had sprung from Black churches. Black Banks made limited investments in other Black enterprises. Irish immigrants in American cities organized many building and loan associations to provide capital for home construction and purchase. They, in turn, provided work for many Irish home-Building contractor firms. Other ethnic and minority groups followed similar practices in founding ethnic-directed financial institutions.

The sociological findings do not prove following

- 1. Communities are more assured when they deal with people from their own race or emigrant nations
- 2. There is rampant ethnic and racial bias in the investment sector
- 3. Social hyper cross pollination had not begun at the time of these studies
- 4. Ways and means of approaching resources are limited for marginal groups.

****Answer:**** C

Question 21

– Read the given passage and answer the questions that follow

A small pole about thirty centimeters long stuck up from a square box on the lab table. There was a ring, only a few centimeters across, encircling the base of the pole. The strange item would have looked a bit like a carnival ring-toss game were it not for the cord snaking out of the side of the box and the single toggle switch on top.

“What the heck’s this?” Kusanagi muttered, bending down to stare at the device.

“I wouldn’t touch that if I were you,” warned Kishitani beside him.

“Nah, I’m sure it’s fine. If this thing were dangerous, he never would’ve left it lying out here like

this.” Kusanagi flicked the switch. The moment he did so, the ring around the pole began to float upward.

Kusanagi gaped. The ring was hovering in mid-air, wobbling slightly.

“Try pushing the ring down,” came a voice from behind him.

Kusanagi looked around as Yukawa walked into the room, a file folder cradled in his arms

“Welcome back. Class?” Kusanagi asked as he turned back to the floating ring.

Reaching out

gingerly, he tried to push it down with his fingertips. A second later he yanked back his hand.

“Yowch! That thing’s hot!”

“It’s true, I’m not in the habit of leaving dangerous objects lying around. I do, however, assume a

basic knowledge of physics.” Yukawa strolled over to the table and flicked off the switch. “This

Device is only high school-level physics, as a matter of fact.”

“Well, I didn’t take physics in high school,” Kusanagi said, blowing on his fingertips. Kishitani

laughed immoderately.

“Who’s your friend?” Yukawa asked with a raised eyebrow.

The smile vanished from Kishitani’s face and he bowed curtly. “Kishitani. I work with Detective

Kusanagi. I’ve heard a lot about you, Professor. You’ve helped us out with a lot of Cases, haven’t you? They call you Detective Galileo down at the station.”

Yukawa frowned and waved his hand. “Don’t call me that. I don’t help out because I enjoy it, you

know. I just couldn't bear listening to Detective Kusanagi's faulty attempts at reasoning on a certain occasion, and I made the mistake of correcting him. I'd be wary of spending too much time with him, if I were you. You might catch whatever he has that's hardened his brain into a rock."

Which option gives an erroneous description of the object that Kusanagi and Kishitani found in Yukawa's lab?

1. The device or the experiment worked only when the single toggle switch on top would be turned on.
2. The device, based on experimentation related to concepts of Physics, had a pole and a Box attached to it.
- 3. The square box on top of the thirty centimeters long pole had a cord attached to one side of it.
- 4. On the base of the pole, a ring encircled the base, which would float upwards if the switch was turned on.

****Answer:**** The square box on top of the thirty centimeters long pole had a cord attached to one side

of it.

Question 22

– On the basis of the narrative style and tone of the passage, what may

Be deduced as the most correct option?

1. The detectives had high regard and admiration for Yukawa's knowledge and even though he was not a part of the official police force, they did not hesitate to take his advice on

Cases from time to time.

- 2. Other than Kusanagi and Kishitani, no one else in the police force knew about Yukawa's involvement in official cases and so, the two wanted to keep it a secret from others.
- 3. Professor Yukawa was a scientist who had high regard for detective work and therefore, whenever he was approached by the official police force to help them in any case, he obliged
- 4. Professor Yukawa encourages Kishitani to follow the principles of detective work of Kusanagi who had become a hardened detective as a result of his experience of several years of work

****Answer:**** Professor Yukawa was a scientist who had high regard for detective work

and therefore, whenever he was approached by the official police force to help them in any case, he obliged

Reasoning Ability

Question 1

– Which is the wrong term in the following letter cluster ?

GK, HJ, OS, MO, WB, RT, EI, WY

- 1. RT
- 2. WB
- 3. EI
- 4. WY

****Answer:**** WB

Question 2

– Three pair of letter cluster out of four pair of letter cluster are having

same relation, find the one having different relation:

- 1. ROQCTMP : HURTXPS
- 2. XDNYPGL : DAGQSJO
- 3. GCMRPVT : WJFPSYW
- 4. TMIKJUB : PWPLLWE

****Answer:**** Not provided

Question 3

– In a certain code language,

A+ B means 'A is the mother-in-law of B' A

– B means 'A is the mother of B'

Ax B means 'A is the son of B'

A÷ B means 'A is the father of B'

Based on the above, how is P related to G if 'F + P – Q x T X K – G'?

- 1. Mother
- 2. Brother's wife
- 3. Sister
- 4. Wife

****Answer:**** b Brother's wife

Question 4

– In a certain code

P%Q means 'P is the daughter of Q'

P@Q means 'P is the son-in-law of Q'

PxQ means 'Q is the mother of P'

PQ means 'Q is the sister of P'

P\$Q means 'Q is the son of P' and

P*Q means 'Q is the wife of P'

Question 5

– An apple seller is selling apples by packing them in a physical multi

layered box. There are 4 rows and 6 columns in the first layer of the box and each
(missing) layers is 1 more than in both the dimensions of the previous layer.

What will be the number of apples in the fourth layer from the first?

- 1. 63

- 2. 24

- 3. 48

- 4. 35

Question 6

– Given below is a question followed by two statements, I and II each

Containing some information. Decide which of the statements is/are sufficient/
necessary to answer the question.

Question:

What is the code for 'functioning' in a certain language?

Statements:

1. In a certain code language, "Our Laptop not functioning" is coded as "tz jz ez sz" and "Laptop consists Camera and Touchpad" is coded as "rz fz mz jz uz"
2. In a certain language, "Smooth preparation is good" is coded as "lz dz pz gz" and "Smooth functioning is good" is coded as "pz gz dz sz"

- 1. Statement I alone is sufficient while II alone is insufficient
- 2. Either statement I alone or statement II alone is sufficient
- 3. Statement II alone is sufficient while I alone is sufficient
- 4. Both the statements I and II together is necessary.

****Answer:**** D Both the statements I and II together is necessary.

Advanced Coding

Question 1

John is mathematics teacher in one of the classes of Dale's school. The class has N students.

Today is their yearly exam. And John is very strict on the answers.

As per John, no one is allowed to cheat. He will find it out from the final marks of two adjacent students.

The marks of N students are mentioned in $M[]$, with index 1 to N .

All of the students are seated one behind the other based on their roll numbers, which is from 1 to N .

If any two adjacent (one behind the other) students have same marks, then it will be considered as cheating, and the latter student will be removed from the merit list. The student sitting behind is considered as cheater.

At the next level is the process of re-evaluation, which is performed by another Department. The re-evaluation can impact marks of K students. The re-evaluation will update their marks and then again, the 3 final merit list will be declared.

The re-evaluation will be in the format of $X\ Y$. Here 'X' is the index of the student whose marks needs to be changed from the original value to Y . Each re-evaluation is independent, and the merit list is updated after each re-evaluation.

After each re-evaluation the merit list is updated, where the cheater is filtered out if there is any.

Your task is to find the maximum number of students that were part of the merit list After each re-evaluation.

Only the final merit list is visible to all students.

Let us try to understand it with an example.

Consider there are total $N = 5$ student with the respective marks $M = [1, 1, 2, 5, 2]$.

So, $N = 5$ and $M = [1, 1, 2, 5, 2]$.

After re-evaluation there were 2 ($K = 2$) students whose marks are updated. $X[1] = 3$ (Y) $X[4] = 2$ (Y) Hence in our scenario, input will look something like 5 2 1 1 2 5 2 1 3 4 2 Now as per teacher's rule, no two students should have same marks, otherwise they will all be removed from the merit list. Let's consider the first re-evaluation: the marks of $M[1]$ which is currently 1, will now be updated to 3. So, the marks will look something like $= [3, 1, 2, 5, 2]$. Clearly no two adjacent students have same marks, hence all the students who participated will

Be in the merit list. So, the answer is 5. So, the answer is 5 for this re-evaluation. In the second re-evaluation: the marks of $M[4]$ which is currently 5, will now be updated to 2. So, the marks will look something like $= [3, 1, 2, 2, 2]$. Clearly $M[3]$, $M[4]$ and $M[5]$ have all the same marks. But only the students sitting behind will be considered as a cheater. So, $M[4]$ and $M[5]$ will be taken out from the merit list. So, now the merit list

Consists of only 3 students $M[1]$, $M[2]$ and $M[3]$. So, the answer is 3 for this re-evaluation.

Python

```
Def fun(x, y, n): Arr[x - 1] = y
```

```
ans = 1
```

```
for i in range(1, n):
```

```
if arr[i] != arr[i - 1]:
```

```
ans += 1
```

```
return ans
```

```
n = int(input())
```

```

k = int(input())
Arr = []
for i in range(n):
    Arr.append(int(input()))
for i in range(k):
    x = int(input())
    y = int(input())
    print(fun(x, y, n), end="")

```

Question 2

Bob is going to bet today on horse riding. There are N horses listed in a sequence of 1 to N.

The probability of winning of each horse is different so the prices for making a bet on the horses are not the same. There is no limit on the number of horses on which he can bet, but he thinks that if he bets on a continuous sequence of horses then he has a better chance to win. Bob will get K units of money if any horse on which he bets will win. But as the award is only K units so he wants to put money less than K.

Bob wants to bet on as many horses as he can. As you are his best friend, he reached out to you for help, can you please find the length of the maximum continuous sequence of horses on which Bob can make a bet, and remember he will invest money less than K units. If there are more than one possible combinations,

Bob will bet randomly on any one of them. Given the number of horses(N), reward money(K), and price of betting on N horses in order. Hint: For each starting index of a horse, its end index in sequences will be equal to or greater than the end index of the previous starting index. Example 1: • Input: 90 100 -> N = 10, K=100 30 40 50 20 20 10 90 10 10 10 -> Price to make bet on each horse in order • Output: 3 • Explanation: There are 10 horses, and the reward money is 100. So, Bob will put money less than 100. There are two possible o sequences of length three whose total money for betting is less than 100, i.e. [50 20 20]

(sum is 90) and [10 10 10] (sum is 30). Bob will choose randomly one sequence from these two. As none of the other sequences with length greater than 3 will have a price less than 100 so the answer will be 3. Example 2: • Input: 10 100 -> N = 10, K=100 10 90 80 20 90 60 40 60 70 75 -> Price to make bet on each horse in order • Output: 1 • Explanation: There are no two consecutive horses for which the sum of price is less than 100. So, Bob will choose randomly any one horse. And the max length of the sequence will be 1.

Constraints: • $2 \leq N \leq 105$ • $1 \leq K \leq 109$ • $1 \leq A_1, A_3 \dots A_N \leq 109$ The Input format for testing: The candidate has to write the code to accept 2 inputs. • First Input: It will contain two integer N (number of horses) and K (reward money) • Second Input: It will contain N integers, each separated C++ #include <bits/stdc++.h> using namespace std; int main() { int n,m; Cin>>n>>m; int a[n][m]; int count=0; for(int i=0; i<n;i++){ for(int j=0; j<m; j++){ Cin>>a[i][j]; } } for(int i=0; i<n;i++){ for(int j=0; j<m; j++){ if(a[i][j]==a[i+1][j]) Continue; else{ if(a[i+1][j]==1 && (a[i+2][j]==1)) Count++; } } Cout<< count<<"\n"; return 0; } Python n, m = map(int, input().split()) A= [] for i in range(n): Aappend(list(map(int, input().split())))

ans = 0

for i in range(n-2):

for j in range(m):

if a[i][j] == a[i + 1][j]:

Continue

elif a[i + 1][j] == 1 and a[i + 2][j] == 1:

ans += 1

print(ans)

Advanced Quantitative

Question 1

– Three quantities x, y, z are respectively in the proportion 3:5:7 . What is

the value of $2x-3y+4z-x-2y+2z-x-2y+2z$ $2x-3y+4z$?

****Answer:**** 19

Question 2

– A number y is drawn arbitrarily from the set $\{1, 2, 3, \dots, 120\}$. The

probability that $(y+120y)^{>23}(y+y120)^{>23}$ is:

- 1. 0.78
- 2. 0.85
- 3. 0.93
- 4. 0.91

****Answer:**** 0.93

Question 3

– Find the length of the chord defined by $y = 1 + 2x$ on the circle

$$x^2+y^2=5$$

- 1. 34304330
- 2. 45305430
- 3. 56306530
- 4. 23303230

****Answer:**** 45305430

Question 4

– If x , y and z are positive integers such

that $x^8=y^6$, $x^3=z^5$ and $y^{27}=z^m$, then what is the value of m ?

****Answer:**** 60

Question 5

– If α and β are the roots of quadratic

equation $px^2+qx+r=0$ then $(\alpha^5\beta^8+\alpha^8\beta^5)$ is equal to

- 1. $qr^5p^8(3pr-q^2)p^8qr^5(3pr-q^2)$
- 2. $qr^5p^6(3pr-q^3)p^6qr^5(3pr-q^3)$
- 3. $qr^5p^7(3pr-q^3)p^7qr^5(3pr-q^3)$
- 4. $qr^4p^8(3pr-q^2)p^8qr^4(3pr-q^2)$

****Answer:**** $qr^5p^8(3pr-q^2)p^8qr^5(3pr-q^2)$

Question 6

– The value of k satisfies the inequation $|k-12|\geq k^2-9k+12$

$12| \geq k^2 - 9k + 12$ are in the range $[a, b]$. Then find the values of a and b .

- 1. $a = 1, b = 7$
- 2. $a = 0, b = 8$
- 3. $a = 1, b = 8$
- 4. $a = 0, b = 7$

****Answer:**** $a = 0, b = 8$

Question 7

– A five-letter password is to be formed using letters from the alphabet

exactly once. If the first letter is not Z and the last is not A and the letter in the 4th place is alphabetically next to the letter in the first place of the password, then how many possible such passwords are there?

****Answer:**** 3,03,600

Advanced Reasoning

Question 1

– Read the given information carefully and answer the question that

follows

The Government has started a new scheme under which thousands of houses have been constructed for the families of different income levels. More than 12000

houses are vacant at various places in the state. A lot of people in the state are homeless but the applications received for the government scheme house allotment are less than 2000. The Government wants to allot houses to the needy people and for that a study has been conducted which reveals that there are certain Discrepancies in the scheme. To address the challenges, the Government has decided to take over the vacant flats and allot them to Government servants only. Is the Government's decision right?

Enter the answer in the space provided on the screen. Answer should be in whole numbers without any symbols (e.g. 9%, kms, ms, etc.)

****Answer:**** No

Question 2

– A Queen sent Sevens servant name P,Q,R,S,T,U and W to market, buying

some vegetables, U bought vegetables before P but not S. R is the last one to bought vegetable. T bought immediately after P and is subsequently followed by W. Among the servants, who bought vegetable before U?

****Answer:**** S

Question 3

– In a certain code DEAF is written as 3567 and FILE is written as 7465.

How is LIED written in that code?

****Answer:**** 6453

Question 4

– In a random exhibition, seven weavers displayed seven different

varieties namely Chanderi, Banaras, Kanchivaram, Uppada, Pochampally, Chettinad and Kuppadam in each store, which are in a row facing north. Chanderi store was immediate right of Kuppadam store. Kuppadam store was fourth to the right of Kanchivaram. Uppada store was between Banaras store and Chettinad store.

Kanchivaram store, which was third to the left of Banaras store, was at one of the ends. Which stores are on the immediate either sides of Chanderi?

****Answer:**** Kuppadam and Pochampally

Question 5

– A question is given, followed by two statements labelled I and II.

Identify which of the statements is/are sufficient to answer the question.

Question:-

Four sisters – A, B, C and D attended a family get together wearing different variety sarees namely – Kaanchevaram, Dharmavaram, Chendheri, and Pochampally but not necessarily in the same order. A wore Kaanchevaram saree and D wore neither Pochampally nor Dharmavaram saree. Who among them wore Pochampally saree?

Statement:-

1. B does not wear Pochampally saree
2. C neither wore Chendheri nor Dharmavaram saree

- 1. Data in statement II alone is sufficient to answer the question, while the data in statement I alone is not sufficient to answer the question.

- 2. Data in both statements I and II together are not sufficient to answer the question.
- 3. Data in statement I alone is sufficient to answer the question, while the data in statement II alone is not sufficient to answer the question.
- 4. Data either in statement I alone or in statement II alone is sufficient to answer the question.

****Answer:**** Data either in statement I alone or in statement II alone is sufficient to answer the

question.

Question 6

– Read the following information carefully and answer the question given

Below.

A company has following Gratuity (G) and Provident Fund (PF) rules:

1. An employee must have completed one year service to be eligible for either G or PF.
2. An employee resigning or retiring or retrenched after ten years' service gets both G or PF.
3. An employee retrenched or retiring after 5 years but before 10 years' of service gets both G and PF, but that resigning during this period gets either G or PF.
4. An employee retrenched or retiring before 5 years' service gets PF but not G, but that resigning during this period gets neither G or PF.
5. In case an employee dies after 2 years of service, his family gets both G and PF.
6. In case of employee was on leave without pay, such period is deducted from his total years of service and then the above rules are applied.
7. In the case of a lady employee, if she has completed two years' service, two years are added to her actual service before applying the above rules, as a special consideration.

Apply the above rules to the case given below and decide whether the employee is eligible for G and/ or OF. Radhika served for two and half years was retrenched.

- 1. If only G can be given
- 2. If neither G nor PF can be given
- 3. If only PF can be given
- 4. If either G or PF can be given

****Answer:**** If only PF can be given

Question 7

– Read the statements and conclusions carefully. Assuming that the

information given in the statements is true, even if it appears to be at variance with Commonly known facts, decide which of the given conclusions logically follow(s) from the statements.

Statements:

1. Some jeans are skirt
2. Some skirts are saree
3. All saree are jacket
4. All skirt are t-shirt.

Conclusions:-

1. Some t-shirts are saree
2. Some jeans are jacket
3. Some jackets are skirt

4. All saree are jeans

- 1. Neither conclusion I nor II follows
- 2. Only conclusion I follows
- 3. Both conclusion I and III follows
- 4. Only conclusion III and IV follows

****Answer:**** Both conclusion I and III follows

Question 8

– In a code language, 'SHORTHAND' is coded as "LSHSGIWMZ". How

would PUBLISHED be coded in that language?

- 1. WVSYFKHRO
- 2. KYFOHRSWV
- 3. SVWORHKFY
- 4. YFKHROWVS

****Answer:**** YFKHROWVS

Question 9

– Read the following information carefully and answer the question given

Below:-

Chetan, Edwin, Manish, Bishsesh and Amar are friends studying in IIT Delhi. Their

hometowns are Ernakulum, Mangalore, Ahmedabad, Bilaspur and Chandigarh but

not in that order. They are doing their graduation in five different disciplines, Computer science, agriculture engineering, mechanical engineering, biotechnology and electrical engineering, again not in the order of their names stated above.

1. Their names, hometowns and disciplines do not start with same letter
2. Neither Edwin nor Manish is in agriculture engineering and they do not belong to either Bilaspur or Chandigarh.
3. Chetan is neither in mechanical engineering nor in biotechnology
4. Ernakulum is not Bishesh's hometown.
5. The student who is from Bilaspur is in the discipline of mechanical engineering.
6. The student studying agriculture engineering, hometown is Ernakulum, while the student studying biotechnology does not hail from Mangalore.

Who is in the electrical engineering discipline?

- 1. Edwin
- 2. Amar
- 3. Chetan
- 4. Bishesh

****Answer:**** Bishesh

Question 10

– Read the statements and conclusions carefully. Assuming that the

information given in the statements is true, even if it appears to be at variance with

Commonly known facts, decide which of the given conclusion logically follow(s) from the statements.

Statements:

1. Some bees are flies
2. All flies are rhythm
3. Some ants are dragonflies
4. All fleas are dragonflies

Conclusions:-

1. Some dragonflies are bees
2. Some bees are ants
3. Some flies are dragonflies

- 1. Both conclusions I and II follow
- 2. Only conclusion I follows
- 3. Neither conclusion I nor II nor III follows
- 4. Only conclusion II follows

****Answer:**** Neither conclusion I nor II nor III follows

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